

CCA103 SAD

Investigating System Requirements

Chapter 2

1

Chapter 2: Outline

Part I: System Analysis Activities

Part II: What Are the Requirements

Part III: Information Gathering Techniques

Part IV: Activity Diagrams

Learning Objectives

- Describe the activities of systems analysis
- Classify requirements correctly as business, user, functional, or nonfunctional requirements
- Identify and understand different kinds of stakeholders and their contributions to requirements definition
- Employ the requirement elicitation techniques of interviews, JAD sessions, questionnaires, document analysis, and observation
- Develop UML activity diagrams to model workflows

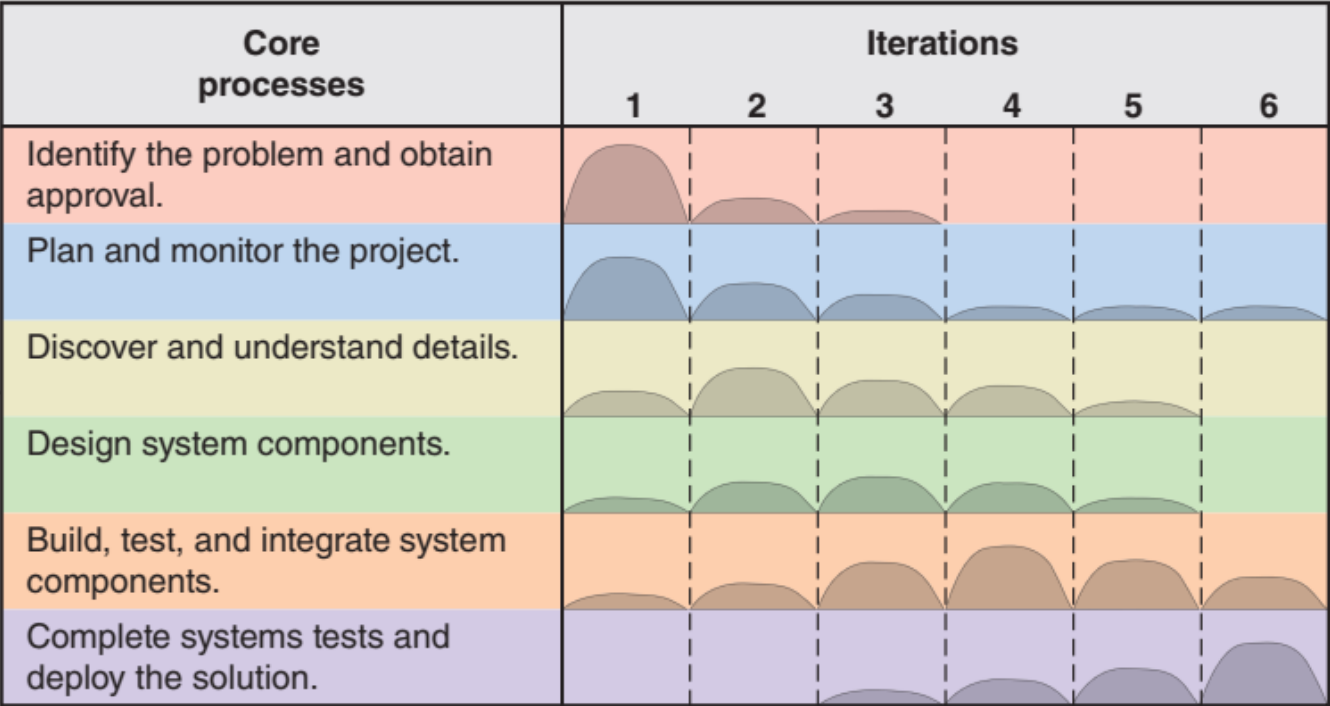
Systems Analysis Activities

Discovery and understanding

FIGURE 2-2 “Analysis” activities from Core Process 3

Analysis activities

Gather detailed information.
Define requirements.
Prioritize requirements.
Develop user-interface dialogs.
Evaluate requirements with users.



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By completing these activities, the analyst defines in detail what the information system needs to accomplish to provide the organization with the desired benefits

Gather Detailed Information

Gather information from users directly through interviews and observation of their daily work.

Engage with all relevant stakeholders, including current users, future users, and those familiar with similar systems.

Review existing materials such as planning documents, policy statements, and current system documentation.

Study similar solutions used by other organizations or vendors facing comparable business needs.

Understand the full work environment, including user activities, work locations, and system interfaces inside and outside the organization.

Develop strong business knowledge, as successful systems analysts must become experts in the business area they support.

Define requirements

- Define system requirements based on information collected from users and existing documents.
- Identify functional requirements, which describe what the system must do.
- Identify nonfunctional requirements, such as usability, reliability, performance, and security.
- Use models to represent requirements, rather than only listing facts and details.
- Review and refine requirements models continuously with users and stakeholders as new information emerges.

Prioritize requirements

- **Prioritize system requirements** once they are clearly understood.
- **Differentiate essential functions** from those that are desirable but not critical.
- **Use business and user context** to judge which requirements matter most.
- **Manage limited resources** by focusing on what is absolutely necessary.
- **Prevent scope creep** by controlling unnecessary expansion of requirements.
- **Guide project iterations** by implementing high-priority requirements earlier for refinement.

Develop user interface dialogs

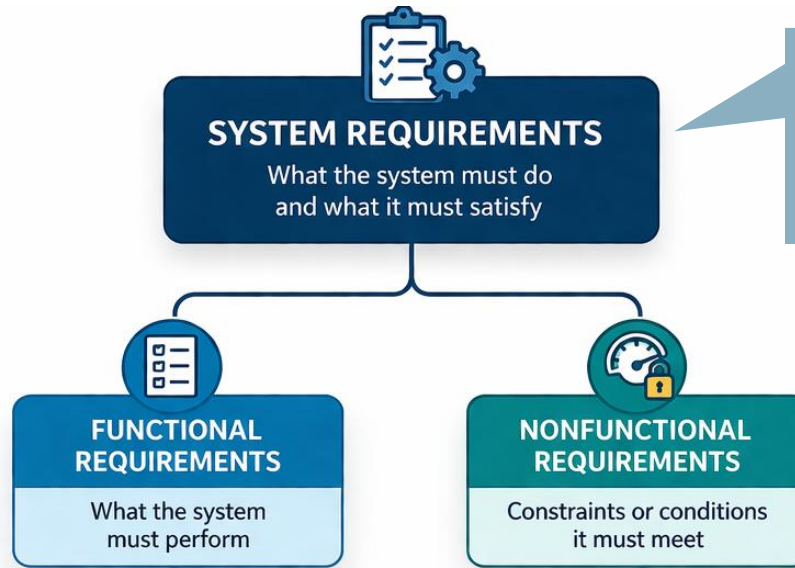
- **Users may be confident or uncertain about requirements**, depending on whether the system replaces an existing one or introduces new functions.
- **Abstract requirements models** such as use cases and activity diagrams can be difficult for users to understand and validate.
- **User-interface validation is easier and more effective** because users can directly see and experience the system.
- **Analysts use interface designs, storyboards, and prototypes** to elicit and document requirements more clearly.
- **Early prototypes can evolve into working system components** through later project iterations.

Evaluate requirements with users

- **Requirements development is iterative**, involving repeated interaction between analysts and users.
- **Analysts gather user input, refine models, and return for validation** as requirements become clearer.
- **Users often have limited time**, so analysts usually complete much of the refinement work independently.
- **Prototypes are useful when paper models are insufficient** or when new technologies need to be tested and visualized.
- The cycle continues until requirements models and prototypes are complete, accurate, and validated by users.

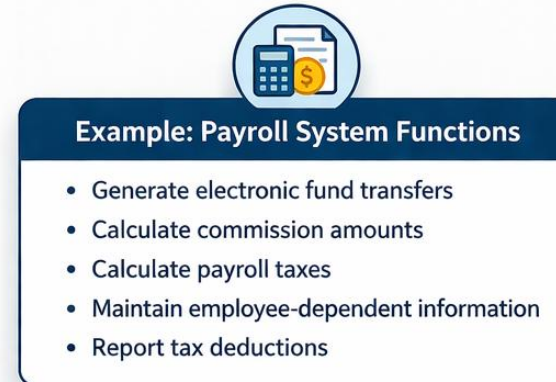
What Are Requirements?

The activities the system must perform to support the users' work



All the activities the new system must perform or support and the constraints that the new system must meet (functional & non-functional)

Required system characteristics other than the activities it must perform or support



The FURPS

- An acronym that stands for **F**unctional, **U**sability, **R**eliability, **P**erformance, and **S**ecurity.

Functional

The activities that the system must perform (i.e., the business uses to which the system will be applied)

Usability

The requirements for operational characteristics related to users, such as the user interface, related work procedures, online help, and documentation

Reliability

The dependability of a system—how often a system exhibits such behaviors as service outages and incorrect processing and how it detects and recovers from those problems.

Performance

Operational characteristics related to measures of workload, such as throughput and response time.

Security

How access to the application will be controlled and how data will be protected during storage and transmission.



- **Functional** - Users can upload photos, like posts, comment, and follow other users.
- **Usability** - The app should have a simple and attractive interface so students can easily find the home page, profile, reels, and messages.
- **Reliability** - The app should not crash when many users are scrolling or uploading stories at the same time.
- **Performance** - Photos, videos, and reels should load within 2 seconds.
- **Security** - Only the account owner should be able to access their private messages, and passwords must be encrypted.

FURPS+

- An extension of FURPS that adds additional categories, including design constraints as well as implementation, system interface, physical, and supportability requirements

1. **Design** constraints describe restrictions to which the hardware and software must adhere
2. **Implementation** requirements describe constraints such as required programming languages and tools, documentation method and level of detail, and a specific communication protocol for distributed components.
3. **Interface** requirements describe interactions among systems.
4. **Physical** requirements describe such characteristics of hardware as size, weight, power consumption, and operating conditions.
5. **Supportability** requirements describe how a system is installed, configured, monitored, and updated.

FURPS+

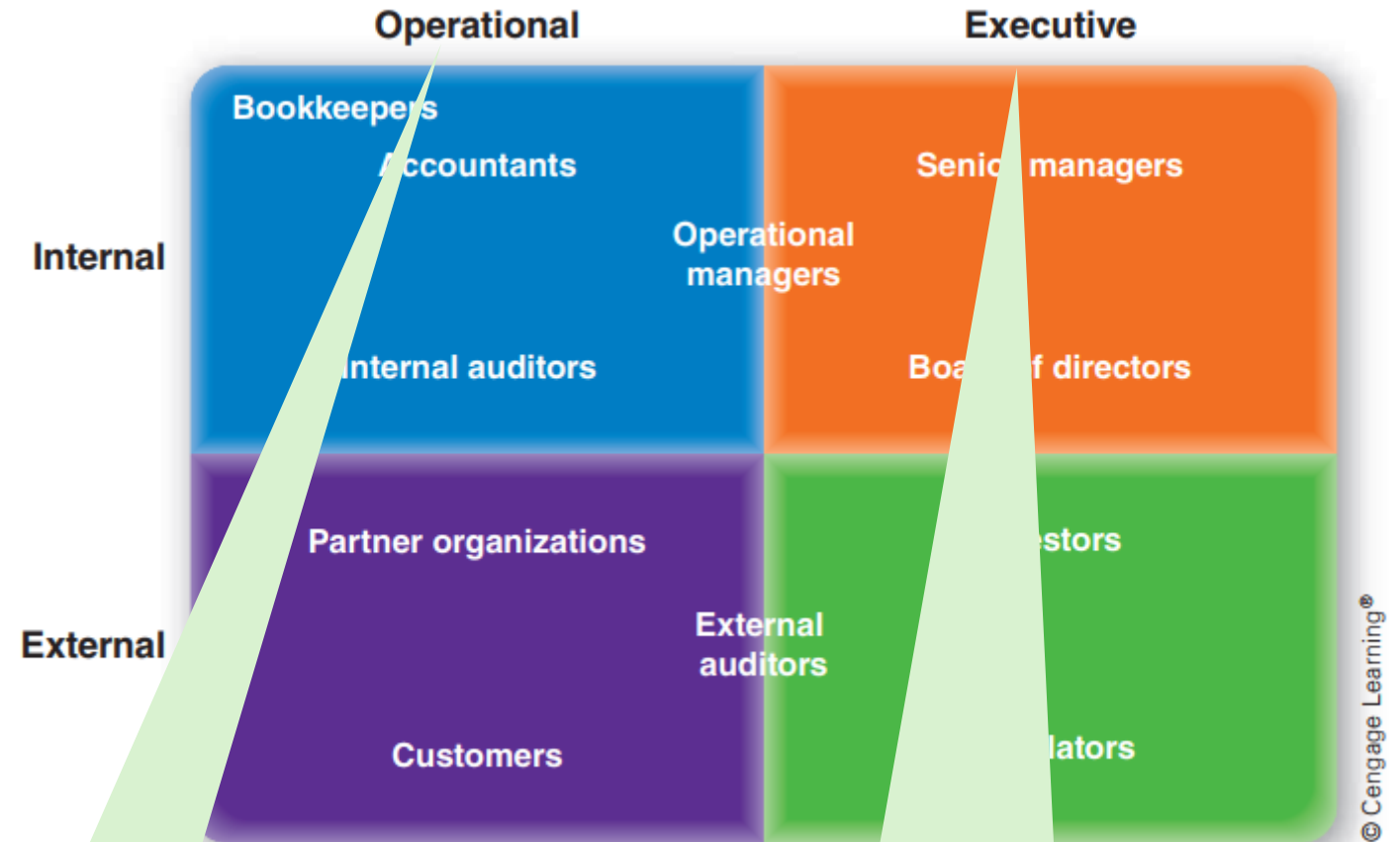
FURPS+ has some gray areas and some overlaps among its categories.

- For example, is a requirement that a battlefield communications device survive immersion in water and operate across a temperature range of –20°C to 50°C a performance or physical requirement?
- Is a restriction to use no more than 100 megabytes of memory a performance or design requirement?
- Is a requirement to secure communication between workstations and servers with 1024-bit encryption a performance, design, or implementation requirement? What is important is that all requirements be identified and precisely stated early in the development process and that inconsistencies or trade-offs among them be resolved.

Stakeholders

persons who have an interest in the successful implementation of the system

- Stakeholders are your **primary source of information** for system requirements.
- Stakeholders are all the people who have an interest in the successful implementation of the system.
- Depending on the nature and scope of the system, this can be a small group, or a large, diverse group.



persons who regularly interact with the system

persons who don't directly interact, but use the information or have a financial interest

Stakeholders: The Client

- The client is the person or group that provides the funding for the project.
- In many cases, the client is senior management.
- However, clients may also be a separate group, such as a corporation's board of directors, executives in a parent company, or the board of regents of a university.
- The project team includes the client in its list of important stakeholders because the team must provide periodic status reviews to the client throughout development.
- The client or a direct representative on a steering or oversight committee also usually approves stages of the project and releases funds.

Information Gathering Techniques

Interviewing
users and other
stakeholders

Distributing and
collecting
questionnaires

**Joint Application
Development
(JAD)**

**Document
Analysis**

Observation

Interview

- Interviewing users and other stakeholders is an effective way to understand business functions and business rules.
- Unfortunately, it is also the most time-consuming and resource-expensive option

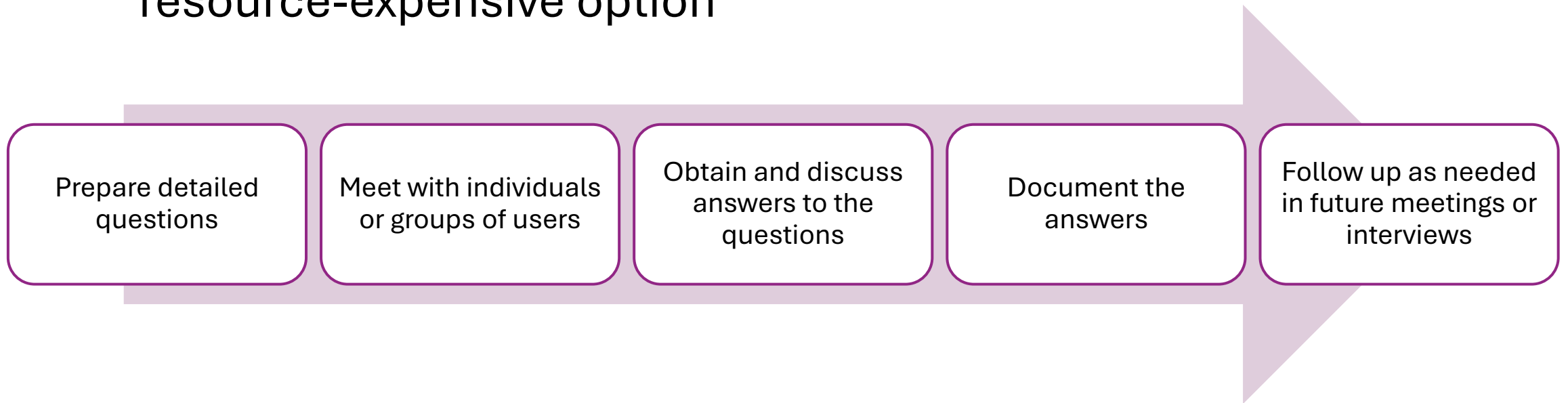


FIGURE 2-6 Themes for information-gathering questions

Theme	Questions to users
What are the business operations and processes?	What do you do?
How should those operations be performed?	How do you do it? What steps do you follow? How could they be done differently?
What information is needed to perform those operations?	What information do you use? What inputs do you use? What outputs do you produce?

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Interview

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Preparing for the Interview



FIGURE 2-8 Sample interview session agenda with follow-up information

Discussion and Interview Agenda
Setting Objective of Interview <i>Determine processing rules for sales commission rates</i> Date, Time, and Location <i>April 21, 2016, at 9:00 a.m. in William McDougal's office</i> User Participants (names and titles/positions) <i>William McDougal, vice president of marketing and sales, and several of his staff</i> Project Team Participants <i>Mary Ellen Green and Jim Williams</i>
Interview/Discussion 1. Who is eligible for sales commissions? 2. What is the basis for commissions? What rates are paid? 3. How is commission for returns handled? 4. Are there special incentives? Contests? Programs based on time? 5. Is there a variable scale for commissions? Are there quotas? 6. What are the exceptions?
Follow-Up Important decisions or answers to questions <i>See attached write-up on commission policies</i> Open items not resolved with assignments for solution <i>See Item numbers 2 and 3 on open items list</i> Date and time of next meeting or follow-up session <i>April 28, 2016, at 9:00 a.m.</i>

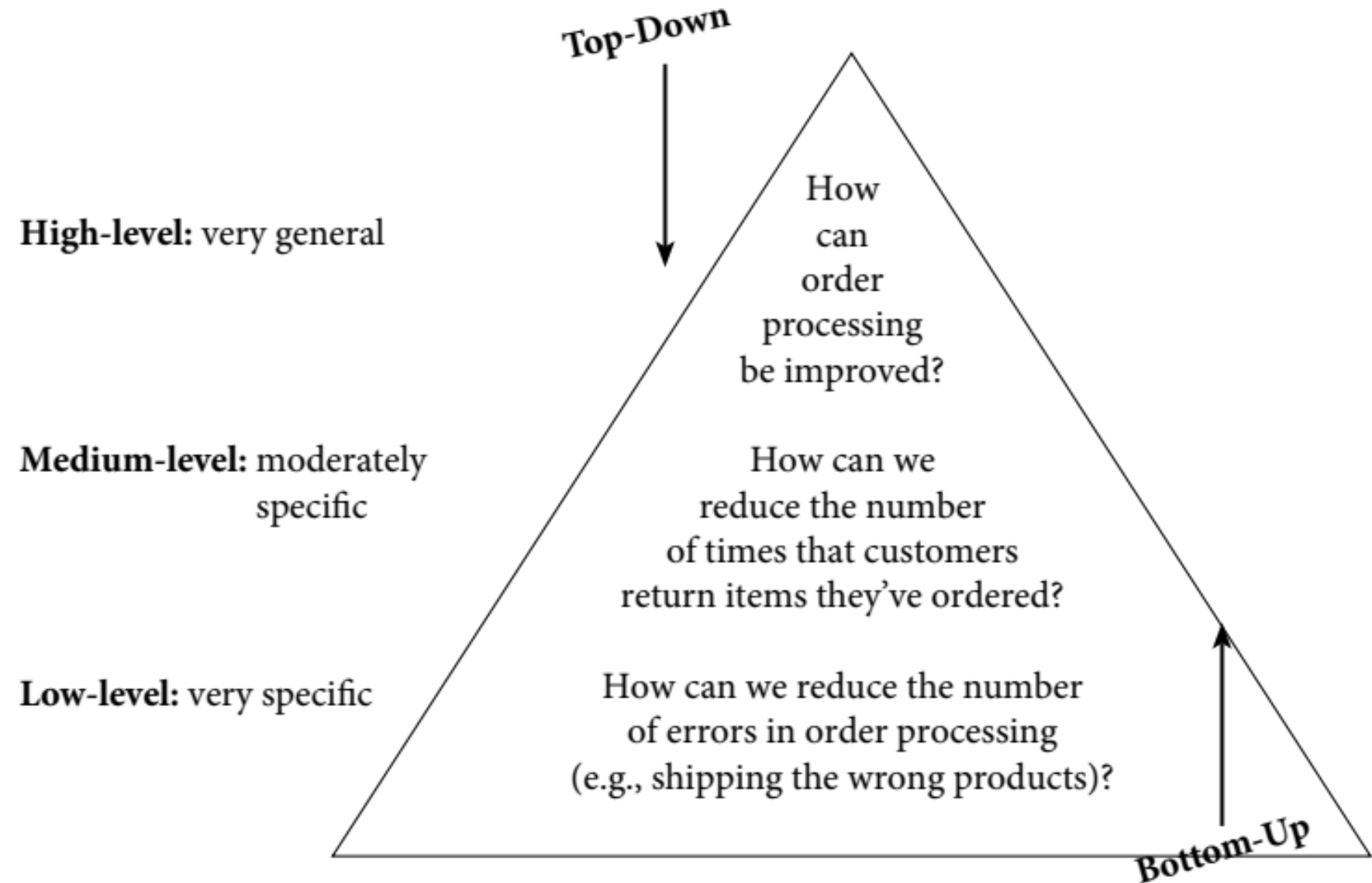
Designing Interview Questions

Types of Questions	Examples
Closed-Ended Questions	• How many telephone orders are received per day? • How do customers place orders? • What information is missing from the monthly sales report?
Open-Ended Questions	• What do you think about the way invoices are currently processed? • What are some of the problems you face on a daily basis? • What are some of the improvements you would like to see in the way invoices are processed?
Probing Questions	• Why? • Can you give me an example? • Can you explain that in a bit more detail?

Typically, interviews include closed-ended, open-ended, and probing questions.

1. **Closed-ended** questions require a specific answer
2. **Open-ended** questions seek a more wide-ranging response from the interviewee
3. **Probing** questions follow up on what has just been discussed so that the interviewer can learn more.

Top-down and bottom- up questioning strategies



Post interview: Interview report

INTERVIEW NOTES APPROVED BY: LINDA ESTEY

Person Interviewed: Linda Estey,

Director, Human Resources

Interviewer: Barbara Wixom

Purpose of Interview:

- Understand reports produced for human resources (HR) by the current system.
- Determine information requirements for future system.

Summary of Interview:

- Sample reports of all current HR reports are attached to this report. The information that is not used and missing information are noted on the reports.
- Two biggest problems with the current system are:
 1. The data are too old. (The HR department needs information within 2 days of month end; currently information is provided to them after a 3-week delay.)
 2. The data are of poor quality. (Often, reports must be reconciled with the HR departmental database.)
- The most common data errors found in the current system include incorrect job-level information and missing salary information.

Open Items:

- Get current employee roster report from Mary Skudrna (extension 4355).
- Verify calculations used to determine vacation time with Mary Skudrna.
- Schedule interview with Jim Wack (extension 2337) regarding the reasons for data quality problems.

Detailed Notes: See attached transcript.

- The report contains interview notes, information that was collected over the course of the interview and is summarized in a useful format.
- In general, the interview report should be **written within 48 hours of the interview**, because the longer you wait, the more likely you are to forget information.
- Often, the interview report is **sent to the interviewee with a request to read it** and inform the analyst of clarifications or updates.
- Make sure the **interviewee is convinced** that you genuinely want his or her corrections to the report.
- Usually, there are few changes, but the need for any significant changes suggests that a second interview will be required.
- Never distribute someone's information without prior approval.

FIGURE 3-8

Interview report.
A template for this
figure is available
on the student website.





Interpersonal skills are those that enable you to develop rapport with others, and they are very important for interviewing. They help you to communicate with others effectively. Some people develop good interpersonal skills at an early age; they simply seem to know how to communicate and interact with others. Other people are less “lucky” and need to work hard to develop their skills.

Interpersonal skills, like most skills, can be learned. Here are some tips:

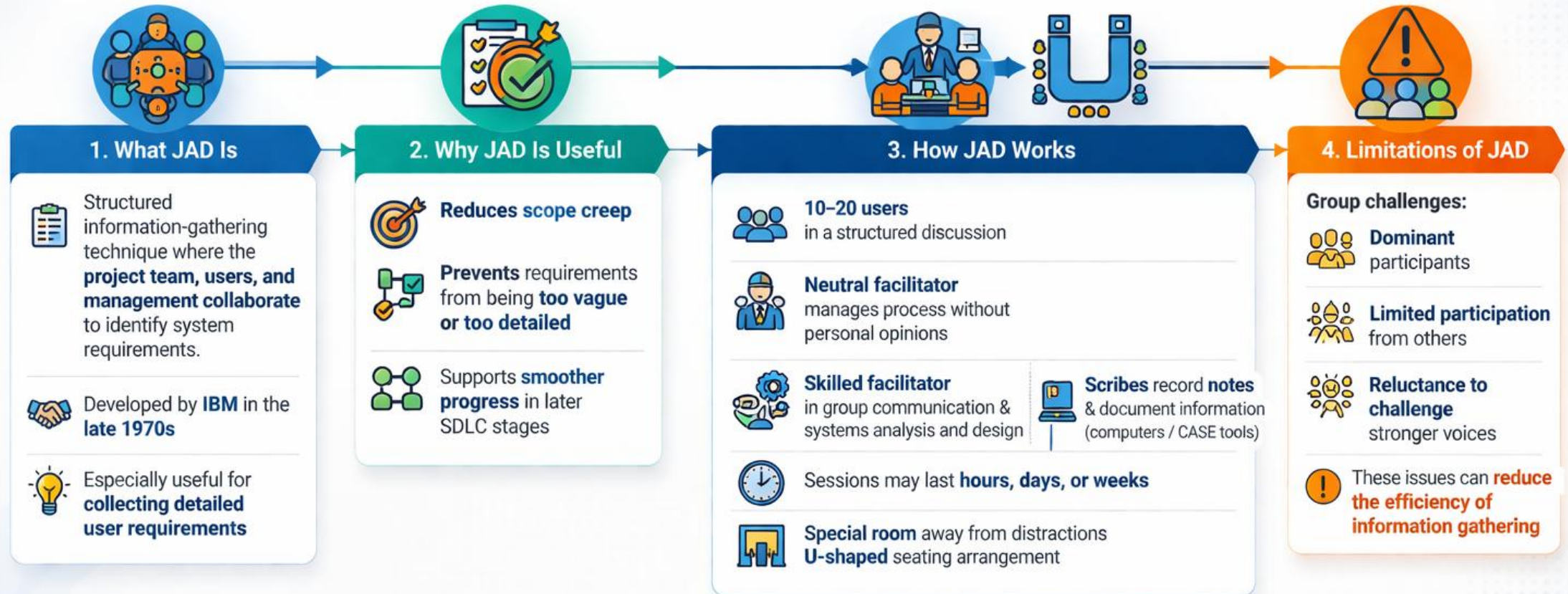
- **Don’t worry, be happy.** Happy people radiate confidence and project their feelings on others. Try interviewing someone while smiling and then interviewing someone else while frowning and see what happens!
- **Pay attention.** Pay attention to what the other person is saying (which is harder than you might think). See how many times you catch yourself with your mind on something other than the conversation at hand.
- **Summarize key points.** At the end of each major theme or idea that someone explains, you should repeat the key

points back to the speaker (e.g., “Let me make sure I understand. The key issues are . . .”). This demonstrates that you consider the information important—and also forces you to pay attention. (You cannot repeat what you did not hear.)

- **Be succinct.** When you speak, be succinct. The goal in interviewing (and in much of life) is to learn, not to impress. The more you speak, the less time you give to others.
- **Be honest.** Answer all questions truthfully, and if you do not know the answer, say so.
- **Watch body language (yours and theirs).** The way a person sits or stands conveys much information. In general, a person who is interested in what you are saying sits or leans forward, makes eye contact, and often touches his or her face. A person leaning away from you or with an arm over the back of a chair is disinterested. Crossed arms indicate defensiveness or uncertainty, while “steepling” (sitting with hands raised in front of the body with fingertips touching) indicates a feeling of superiority.

Joint Application Development (JAD)

A structured technique for **gathering system requirements collaboratively**.



Joint application development meeting room

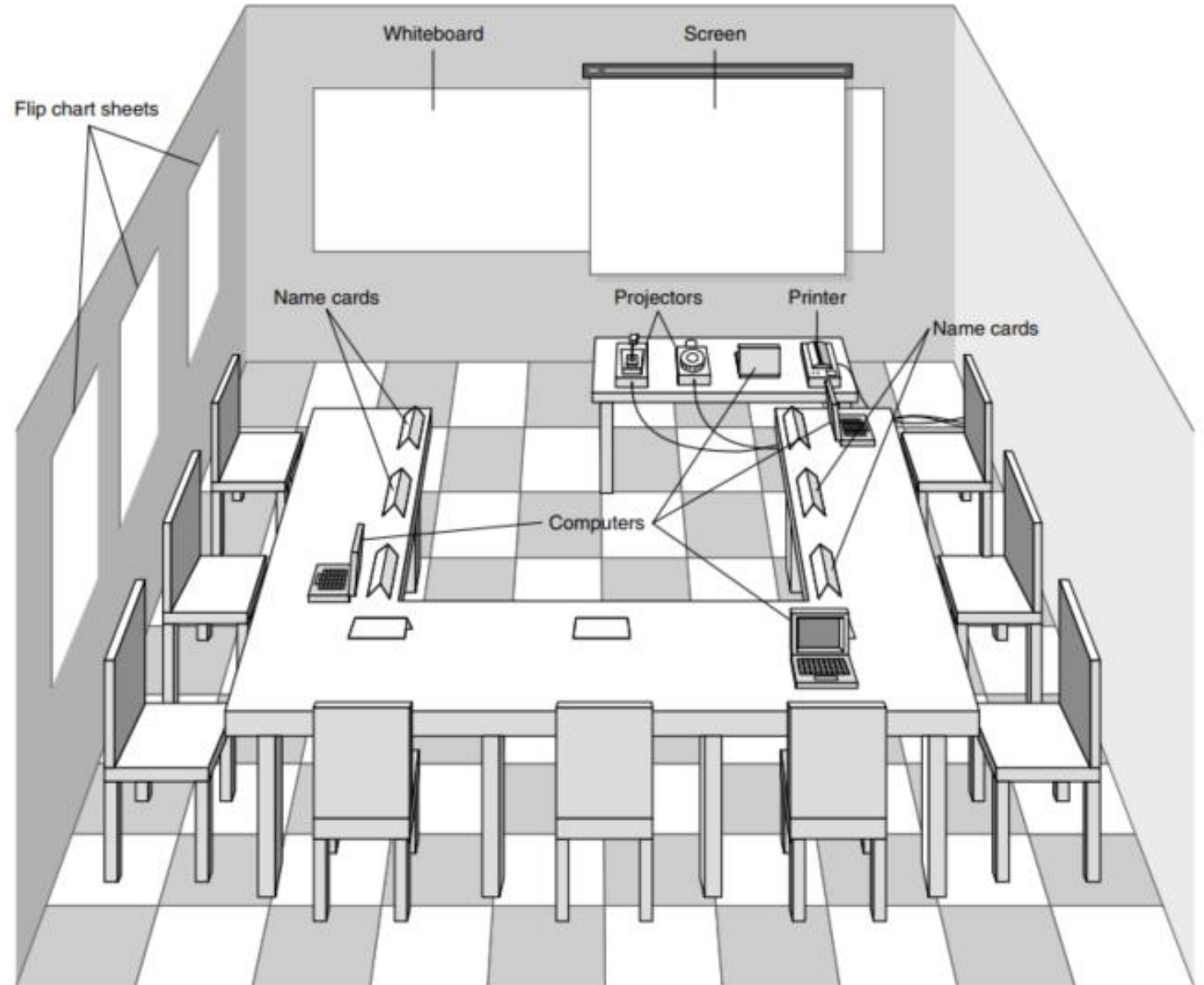


FIGURE 3-9 Joint application development meeting room.

NARM

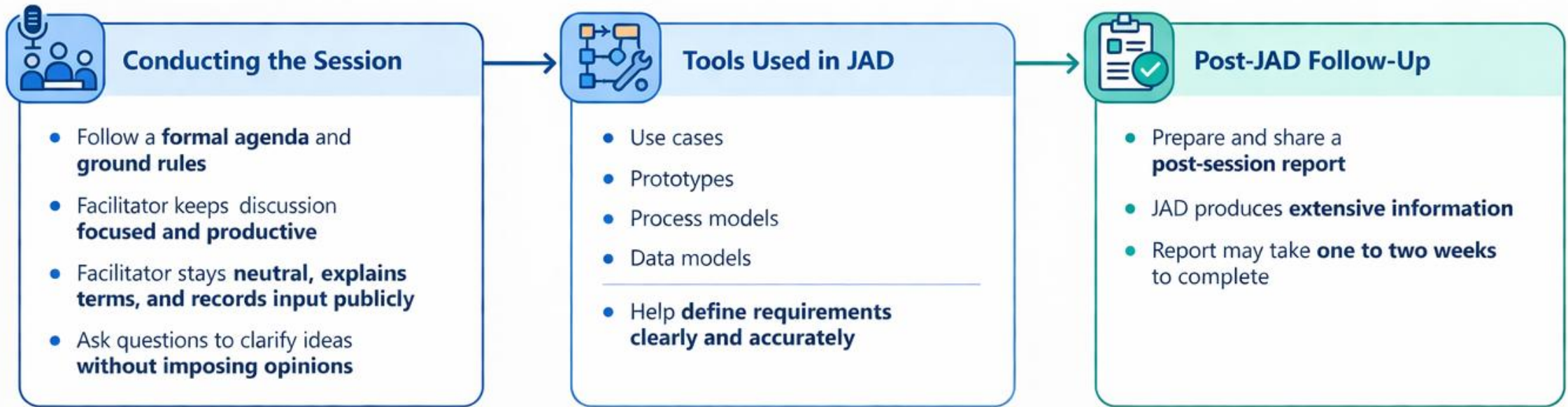
Conducting JAD

Before the JAD Session

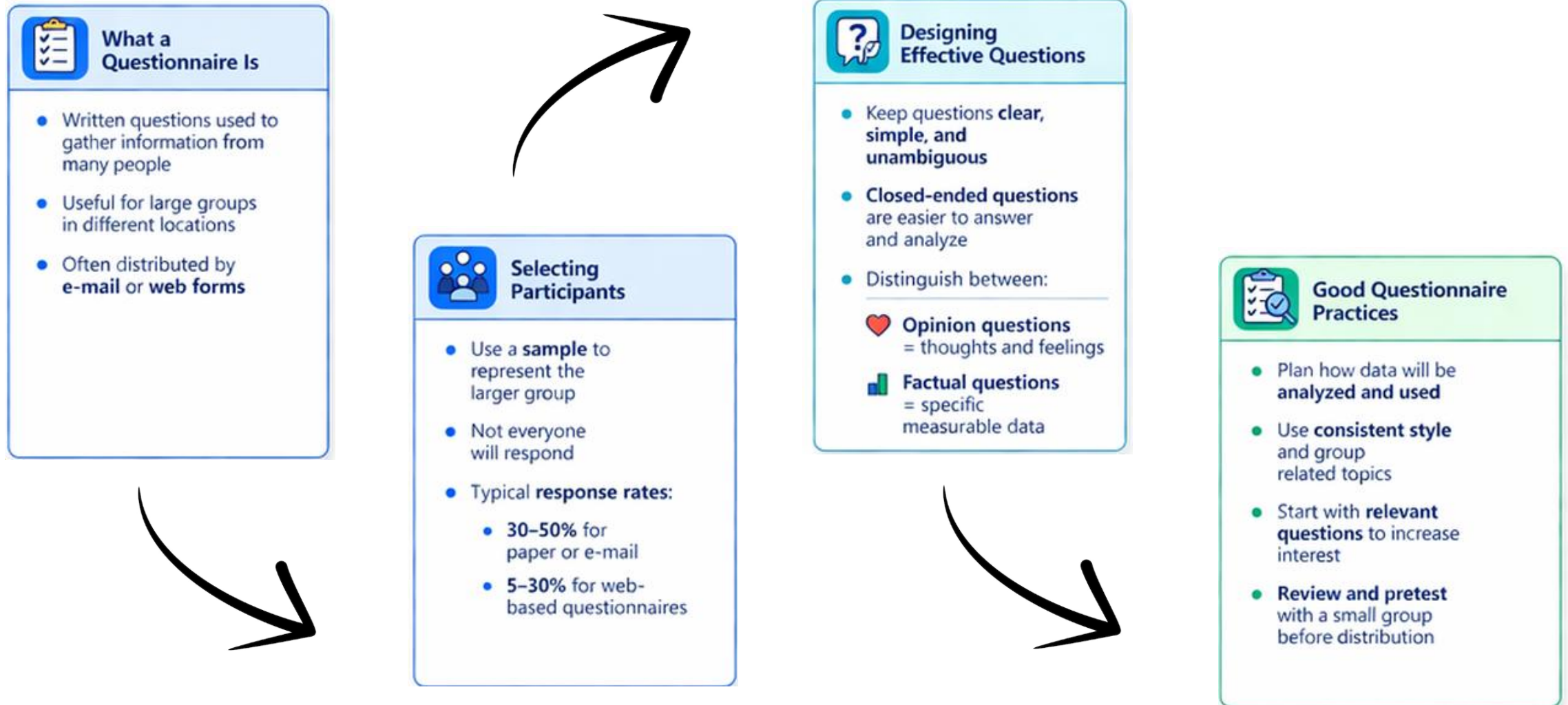


Conducting JAD

During and After the JAD Session



Questionnaire



Questionnaire

Source: Dennis, A. Wixom B.H., Roth R. M. (2021). Systems Analysis and Design (8th ed.). John Wiley & Sons.

FIGURE 3-10
Good questionnaire
design.

- Begin with nonthreatening and interesting questions.
- Group items into logically coherent sections.
- Do not put important items at the very end of the questionnaire.
- Do not crowd a page with too many items.
- Avoid abbreviations.
- Avoid biased or suggestive items or terms.
- Number questions to avoid confusion.
- Pretest the questionnaire to identify confusing questions.
- Provide anonymity to respondents.

FIGURE 4.10

Open-ended questions used for questionnaires.

53. What are the most frequent problems you experience with computer output?

A. _____

B. _____

C. _____

54. Of the problems you listed above, what is the single most troublesome?

55. Why?

Below are questions about yourself. Please fill in the blanks to the best of your ability.

67. How long have you worked for this company?
_____ Years and _____ Months

68. How long have you worked in the same industry?
_____ Years and _____ Months

69. In what other industries have you worked?

Open-ended questions can ask the respondent for lists ...

... or detailed responses ...

... or short answers.

FIGURE 4.11

Closed questions on questionnaires help ensure responses.

Answer questions 23 and 24 by checking the appropriate box.

23. Below are the six software packages currently available. Please check the software package(s) you personally use most frequently.

☐ Microsoft Excel	☐ Microsoft Access
☐ Microsoft PowerPoint	☐ Microsoft Windows Live Mail
☐ Oracle SCM	☒ Visible Analyst

24. "The sales figures are usually late."
☐ Agree ☒ Disagree

Answer questions 25 and 26 by circling the appropriate number.

25. "When the sales figures are prepared by computer data services they are late."

Never 1 Rarely 2 Sometimes 3 Often 4 Always 5

... or circle a number ...

Answer questions 45-48 by circling the appropriate response.

45. The division I am currently in is called

Investments
Operations
Marketing

46. My educational background can best be described as

High School
Some College
Bachelor's Degree
Master's Degree or Higher

My gender is

Male
Female

... or circle the answer itself.

Document Analysis

- Document analysis helps project teams **understand the current (as-is) system by reviewing existing system-related documents.**
- In ideal situations, teams can rely on technical documentation from earlier projects, but many systems are poorly documented or outdated.
- Useful organizational documents include reports, memos, policy manuals, training manuals, organization charts, forms, and user problem reports.
- These documents mainly describe the formal system, but they may not reflect how work is done in practice.
- Differences between the formal system and the informal real system can reveal areas that need improvement, such as unused forms or fields filled in incorrectly.
- Strong evidence for system change appears when users create their own forms, modify existing ones, or combine multiple reports to get the information they need.

CENTRAL VETERINARY CLINIC
Patient Information Card

Name: ~~Buffy~~ Pat Smith

Pet's Name: Buffy *Collie* 7/6/17 Male

Address: 100 Central Court, Apartment 10
Toronto, Ontario K7L 3N6

Phone Number: *416-* 555-3400 *Cell: 416-567-1369*

Do you have insurance: yes

Insurance Company: Pet's Mutual

Policy Number: KA-5493243

The customer made a mistake. This should be labeled **Owner's Name** to prevent confusion.

The staff had to add additional information about the type of animal and the animal's date of birth and gender. This information should be added to the new form in the to-be system.

The customer did not include area code in the phone number. This should be made more clear.

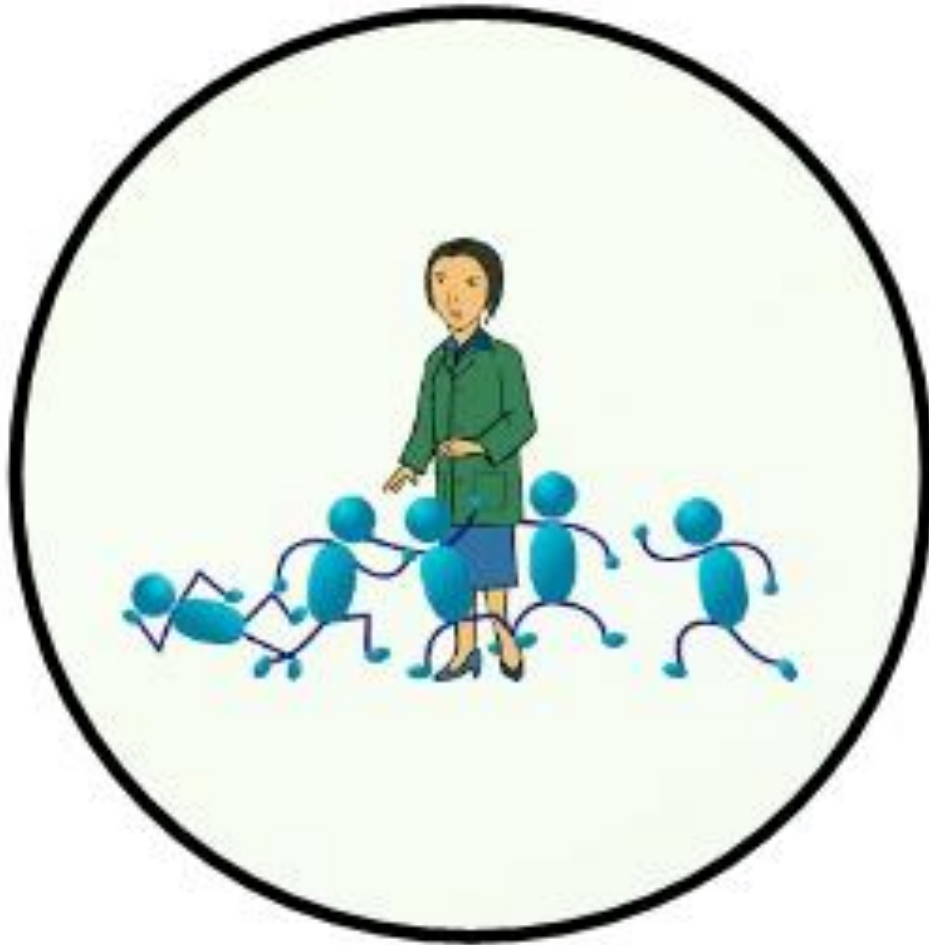
The form does not allow for this fact. It should be added to the to-be system.

FIGURE 3-11
Performing a document analysis.

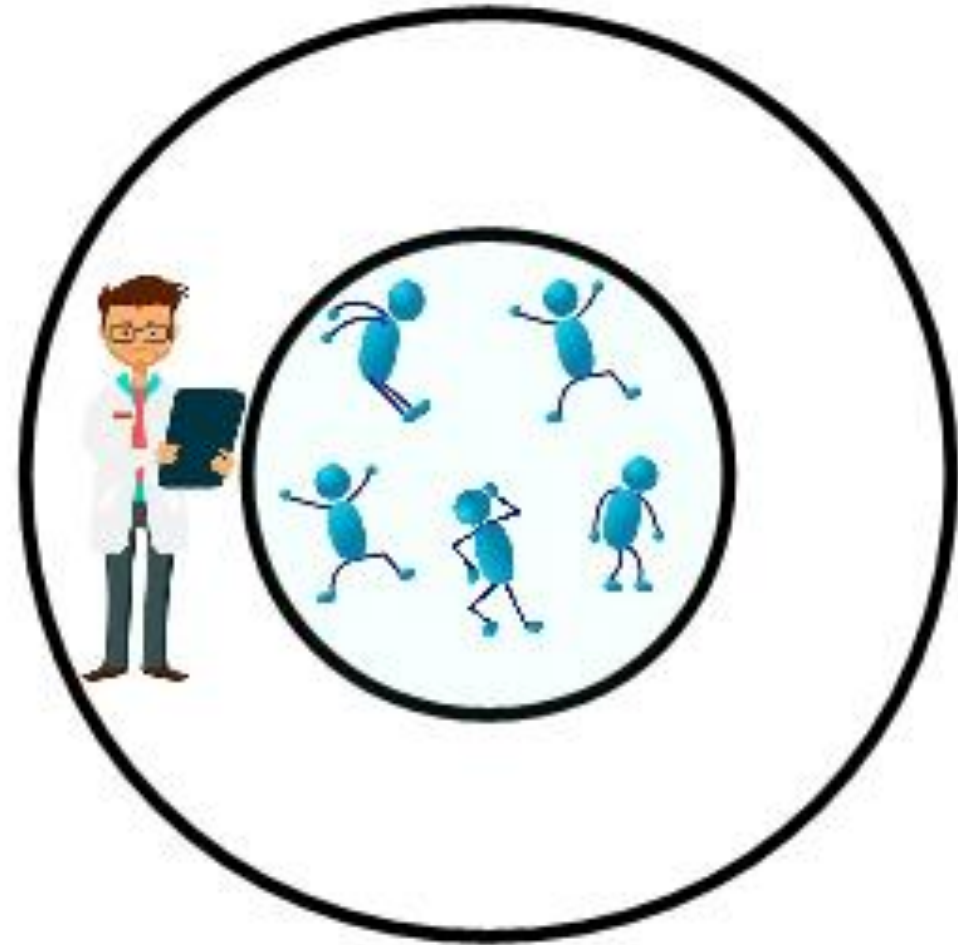
Observation

- Observation is a valuable technique for understanding the **as-is system** by directly watching how work is actually performed.
- It helps analysts see the **real situation**, instead of relying only on what people say in interviews, JAD sessions, or questionnaires.
- Observation is also useful for **verifying information** gathered from other sources.
- Analysts should observe with a **low profile**, so they do not interrupt work or influence people's behavior.
- A limitation is that people often behave more carefully when being watched, so what is observed may not fully reflect their normal routine.
- Observation can reveal clues about roles, power, and system use, and when what is seen does not match what people say, it signals the need for **closer analysis**.

Participant observation



Non-participant observation



Selecting the Appropriate Techniques

	Interviews	Joint Application Design	Questionnaires	Document Analysis	Observation
Type of information	As-is, improvements, to-be	As-is, improvements, to-be	As-is, improvements	As-is	As-is
Depth of information	High	High	Medium	Low	Low
Breadth of information	Low	Medium	High	High	Low
Integration of information	Low	High	Low	Low	Low
User involvement	Medium	High	Low	Low	Low
Cost	Medium	Low–Medium	Low	Low	Low–Medium

FIGURE 3-12 Comparison of requirements elicitation techniques.

Source: Dennis, A. Wixom B.H., Roth R. M. (2021). Systems Analysis and Design (8th ed.). John Wiley & Sons.

Comparing Information-gathering Techniques



Type of Information

- Techniques differ by **analysis stage**
- **Interviews** and **JAD** fit **all 3 stages**
- Document analysis and observation help with the **as-is system**
- Questionnaires support **as-is and improvement ideas**



Depth of Information

- Depth means **richness and detail**
- Interviews and JAD **provide rich detail and explain why**
- Document analysis and observation focus more on **facts**
- Questionnaires give **medium depth with limited reasons**



Breadth of Information

- Breadth means **range of sources and information collected**
- Questionnaires and document analysis cover **many sources easily**
- Interviews and observation **take more time source by source**
- JAD is **moderate** because **many sources meet together**



Integration of Information

- Combining information from **many sources is challenging**
- Different people may provide **conflicting details**
- Resolving differences **takes time and follow-up**
- Rechecking discrepancies may make **users defensive**

Models and Modeling

How do we define requirements?

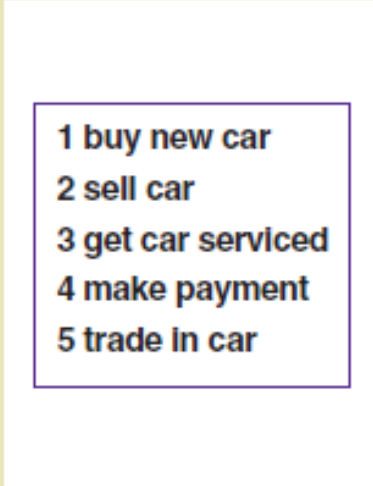
- After collecting information, we create models
- Model is a representation of some aspect of the system being built
- Types of Models
 1. **Textual** model– something written down, described
 2. **Graphical** models– diagram, schematic
 3. **Mathematical** models– formulas, statistics, algorithms
- **Unified Modeling Language (UML)**
 - Standard graphical modeling symbols/terminology used for information systems



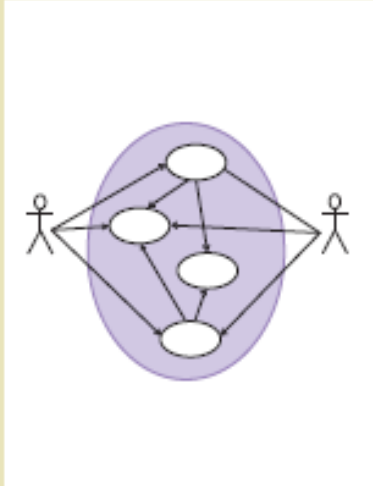
Reasons for Modeling

1. Learning from the modeling process
2. Reducing complexity by abstraction
3. Remembering all the details
4. Communicating with other development team members
5. Communicating with a variety of users and stakeholders
6. Documenting what was done for future maintenance/enhancement

Some
Analysis and
Design
Models



Event list



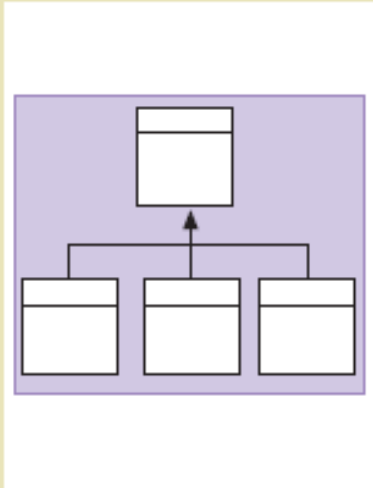
Use case diagram



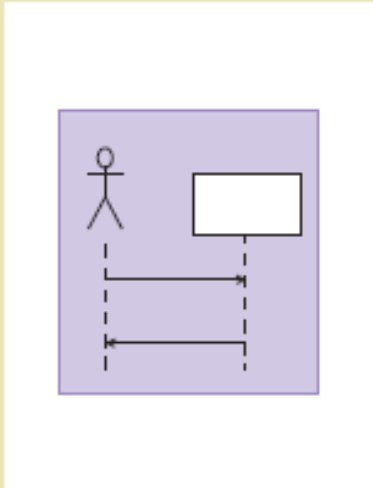
Use case description



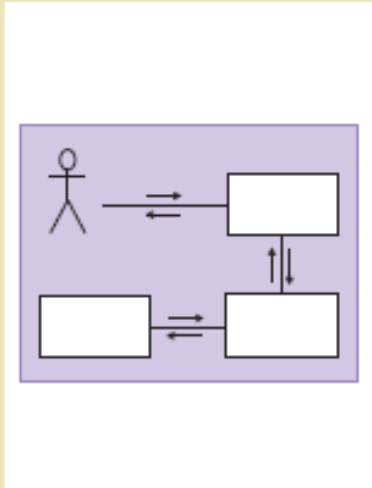
Location diagram



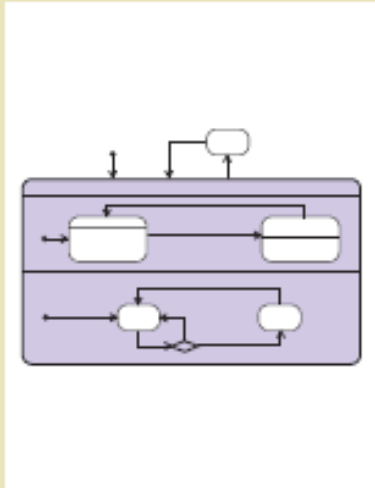
Class diagram



Sequence diagram



Communication diagram



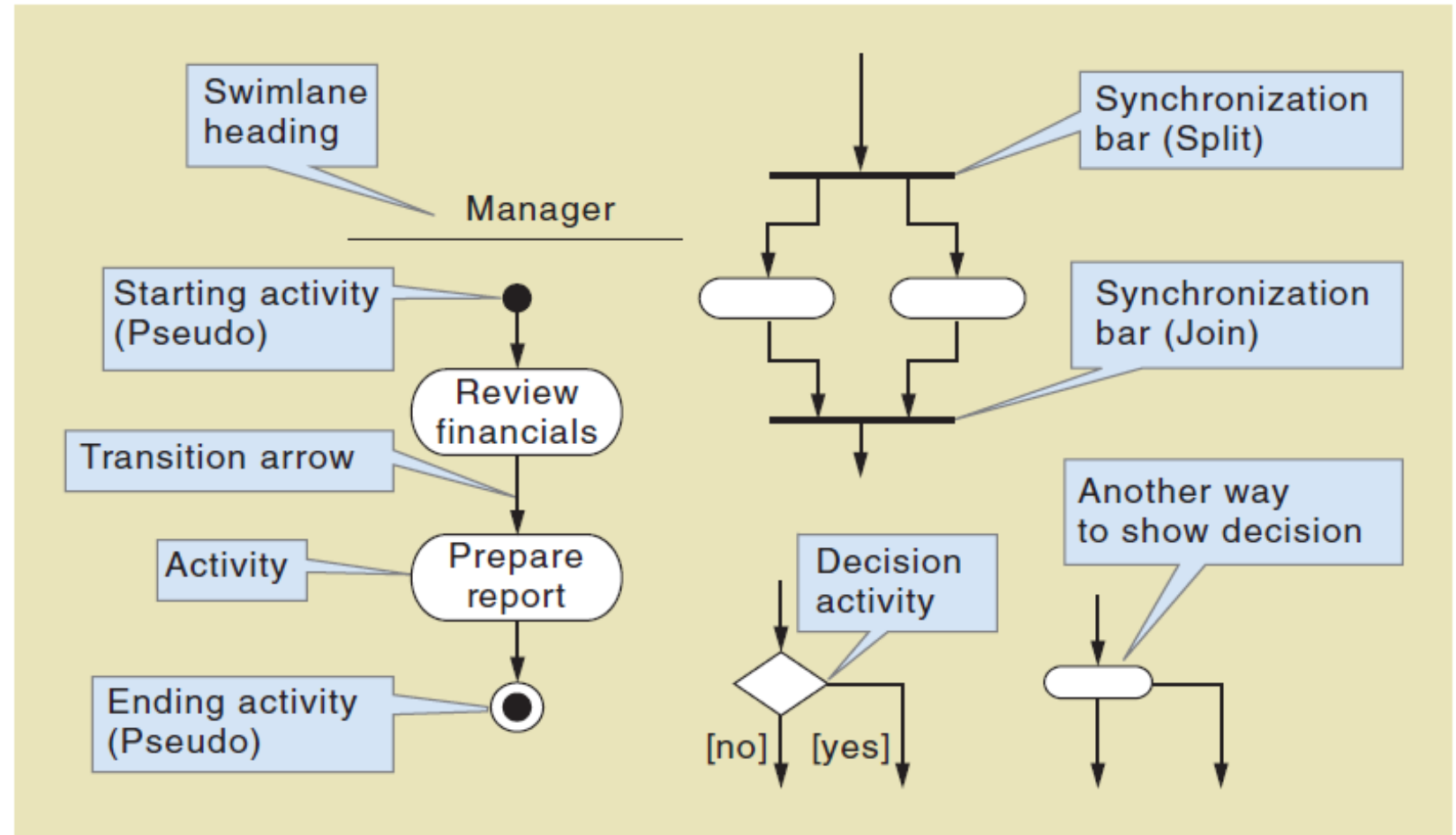
State machine diagram

Documenting Workflows with Activity Diagrams

- **Workflow**
 - sequence of work steps that complete one business transaction or customer request. Workflows may be simple or complex.
 - Complex workflows can be composed of dozens or hundreds of work steps and may include participants from different parts of an organization
- **Activity Diagram**
 - describes user (or system) activities, the person who does each activity, and the sequential flow of these activities
 - Useful for showing a graphical model of a workflow
 - A UML diagram

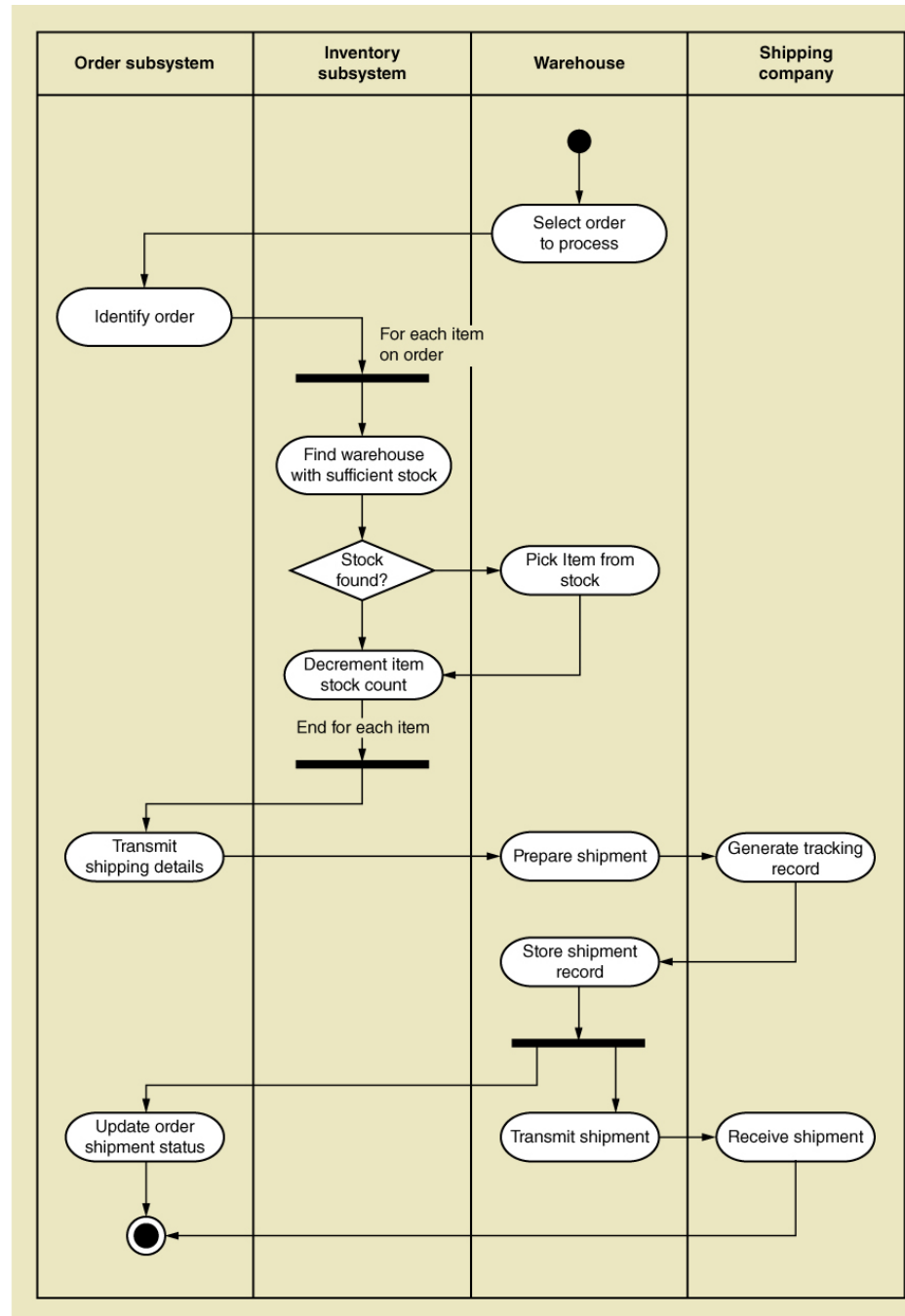


Activity Diagrams Symbols

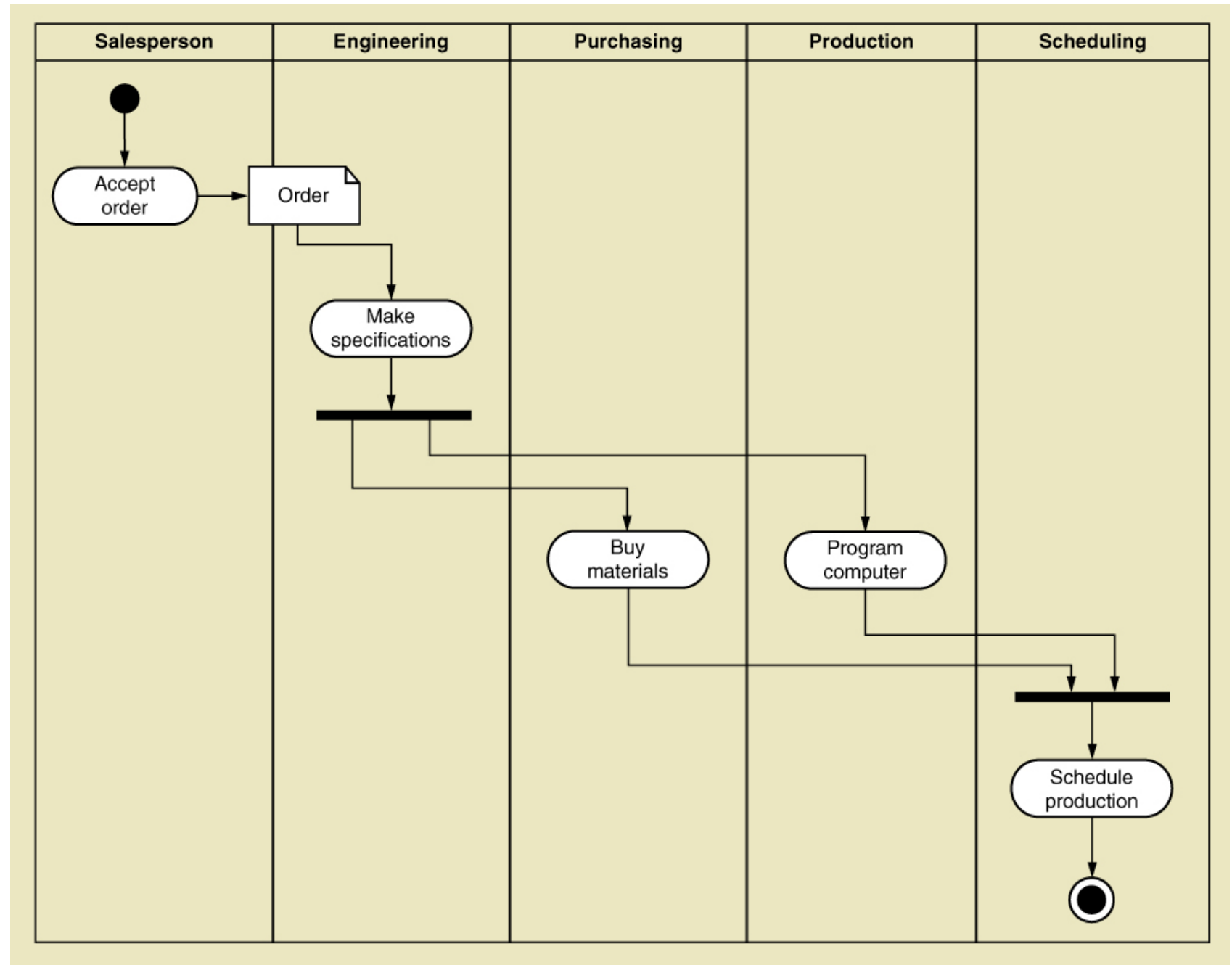


Source: Satzinger, J. W., Jackson, R. B., & Burd, S. D. (2015). Systems Analysis and Design in a Changing World (7th ed.). Course Technology CENGAGE Learning

Activity Diagram for R M O Order Fulfillment



Activity Diagram with Concurrent Paths



References

